🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.EE.8

Write Inequalities

Lesson 7-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

Key Vocabulary **Level 2 Standard**

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x > 5$ means x can be 6, 7, 8, ... — any number greater than 5

Inequality

Write the definition:

$8 > 3$ — eight is larger than three

Greater than

Write the definition:

$2 < 7$ — two is smaller than seven

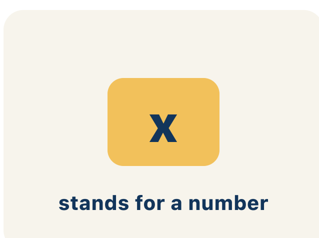
Less than

Write the definition:

'At least 10 players' → $p \ge 10$ (could be 10, 11, 12, ...); 'At most 5 tries' → $t \le 5$ (could be 5, 4, 3, ...)

At least / At most

Write the definition:



In $x > 5$, the letter x stands for any number greater than 5.

Variable

Write the definition:

"No more than 8 people" means people ≤ 8.

No more than

Write the definition:

Guided Notes (Level 2 Standard)

🎯 **WHAT WE'RE LEARNING TODAY**

I can write inequalities to represent real-world situations.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Inequality | Greater than | Less than | At least / At most | Variable | No more than

👆 *Tap any word to see what it means and a picture.*

1. A math sentence comparing values with <, >, ≤, or ≥ is an ______________.

2. The symbol > that means one value is larger than another means ______________.

3. The symbol < that means one value is smaller than another means ______________.

4. 'At least' means greater than or equal to, and 'at most' means ______________ than or equal to.

5. A letter that stands for an unknown number is a ______________.

6. The phrase that means less than or equal to is "no more than," written with the ______________ symbol.

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: Which inequality represents: 'The temperature is less than 40 degrees'?

A. $t < 40$

B. $t > 40$

C. $t \le 40$

D. $t = 40$

Step 1 'Less than' means <, so the inequality is $t < 40$.

✅ **Answer:** A. $t < 40$

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: Which inequality represents: 'You must be at least 48 inches tall to ride'?

A. $h \ge 48$

B. $h > 48$

C. $h < 48$

D. $h \le 48$

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which inequality represents: 'A store can hold no more than 75 people'?

A. $p \leq 75$

B. $p < 75$

C. $p \geq 75$

D. $p > 75$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A clue says the suspect carried at least 12 stolen gems. Which inequality shows the number g of gems?

A. $g \geq 12$

B. $g > 12$

C. $g \leq 12$

D. $g < 12$

Show your work:

2. A witness reports that fewer than 5 people entered the building. Which inequality shows the number n of people?

A. $n < 5$

B. $n \leq 5$

C. $n > 5$

D. $n \geq 5$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write two different real-world situations: one that uses > and one that uses ≥. Explain why the situations need different symbols even though both mean 'bigger.'

Sentence starter: Situation 1 (>): ____. This uses > because ____. Situation 2 (≥): ____. This uses ≥ because ____. The difference is that ≥ includes ____ while > does not.

Show your work:

Reflect — Exit Ticket

Which inequality represents: 'A number n is at least 14'?

A. $n \ge 14$

B. $n > 14$

C. $n \le 14$

D. $n < 14$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Inequality
2. **Notes 2:** Greater than
3. **Notes 3:** Less than
4. **Notes 4:** At least / At most
5. **Notes 5:** Variable
6. **Notes 6:** No more than
7. **We Try (notes):** A. $h \geq 48$ — *'At least' means greater than or equal to, so $h \geq 48$.*
8. **You Try (notes):** A. $p \leq 75$ — *'No more than' means less than or equal to, so $p \leq 75$.*
9. **Try It 1:** A. $g \geq 12$ — *'At least 12' includes 12 and any number above it, so $g \geq 12$.*
10. **Try It 2:** A. $n < 5$ — *'Fewer than 5' does not include 5, so $n < 5$.*
11. **Exit Ticket:** A. $n \geq 14$ — *'At least 14' means 14 or greater, so $n \geq 14$.*